CPE 420: Intro to Data Mining

Project 2

Clustering

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1. Introduction

This project applies clustering techniques to real-world datasets. The main objective is to implement and evaluate unsupervised learning algorithms. The project is divided into multiple tasks, and one main dataset is used, The Asian Religious and Biblical Texts Data Set, which includes two pre-processed Document-Term Matrices (DTM) ,One unlabelled for clustering and one labelled. The implementation is done using Python. Two main files are developed as follows, Project2.ipynb and cluster_analysis.py.

2. Results

Clustering was performed using two unsupervised algorithms, k-means and k-medoid. Ten different cluster sizes k were evaluated, ranging from 2 to 11. The quality of clustering for each k was measured using the Silhouette Coefficient (SC).

Table 1 and Figure 1 show the Silhouette Coefficient for K-Means and K-Medoids.

K	K-Means	K-Medoids
2	0.4231	0.1071
3	0.4284	0.1275
4	0.4238	0.1592
5	0.4246	0.0930
6	0.4264	0.0908
7	0.3882	0.0580
8	0.3874	-0.0829
9	0.3744	-0.0756
10	0.3735	-0.0723
11	0.3647	-0.1333

Tabel 1 : Silhouette Coefficient for K-Means and K-Medoids

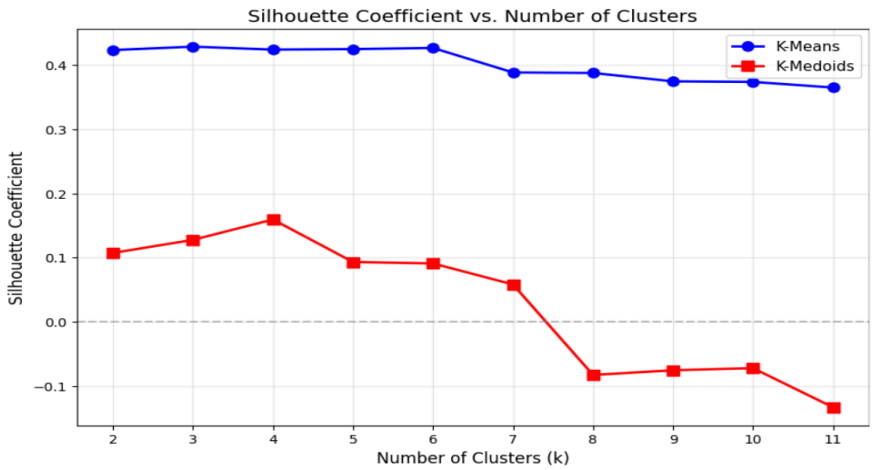

Figure 1 : Silhouette Coefficient for K-Means and K-Medoids

3. Discussion

The best clustering quality was achieved at k = 3 using K-Means, with a Silhouette Coefficient of 0.4284. At this k value, K-Means ran in 0.3921 seconds and K-Medoids ran in 0.2231 seconds. Although K-Medoids was slightly faster, the difference is negligible for this dataset size. K-Means is recommended for text clustering because it consistently produced higher quality clusters across all k values. K-Means achieved an average Silhouette Coefficient of 0.4007 from k=2 to k=11, while K-Medoids averaged only 0.0291. More importantly, K-Means maintained positive SC values for all k, whereas K-Medoids produced negative SC values for $k \geq 8$, indicating arbitrary or incorrect cluster assignments.

The superior performance of K-Means comes from its centroid averaging approach, which handles the sparse, high-dimensional DTM better than K-Medoids' method of selecting actual documents as cluster centers.

Main reasons for poor clustering:

- DTM sparsity and high dimensionality makes distance calculations unreliable
- Vocabulary overlap.
- Short documents lack sufficient term frequency information

In conclusion, K-Means is strongly recommended for text clustering due to its higher Silhouette Coefficients and better stability, while K-Medoids should be avoided for high-dimensional sparse text data.